

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 2

Year: 2014

Subject: 394172 Mathematics II

Section: 15-16

Date: 19 May 2015

Time: 08.00-11.00

Name _____ ID _____ Class _____

Instructions

1. Cheating will result in failure of all classes registered of the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Textbooks are **NOT** allowed.
 4. Calculation devices are **NOT** allowed.
 5. Electronic communication devices are not allowed in the examination room.
 6. This examination has 9 pages (including this page), 15 questions only 12 questions and a total score of 120 points.
 7. Write all your answers in this examination sheet.
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Question 1. Solving the system of equations. (10 points)

$$7x + 5y - 7z = -8$$

$$4x + 2y - 3z = 0$$

$$5x - 4y + 4z = 35$$

Question 2. Solving the system of three nonlinear equations. (10 points)

$$(x + y)^2 - z^2 = 65$$

$$x^2 - (y + z)^2 = 13$$

$$x + y - z = 5$$

Question 3. Given the algebraic equation $\frac{3x}{(x-2)^2} = \frac{2}{x^2-5x+6}$.

Find all solutions. (10 points)

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Question 4. Given a, x, y are integer numbers, suppose that $a \mid (-4x+3y)$ and $a \mid (3x-2y)$. Show that $a \mid x$. (10 points)

Question 5. Given two positive integers $a = 942$ and $b = 480$, find the least common multiple of a and b ($\text{lcm}(a, b)$) and find the greatest common divisor of a and b ($\text{gcd}(a, b)$). (10 points)

Question 6. To solve the absolute value inequality $2|5-2x|-15 \geq 5$ and graph the solution set. (10 points)

Question 7. Solving the absolutely inequality $\frac{|3x-2|}{|x+1|-1} > 2$. (10 points)

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Question 8. Evaluate the value of the radical expression $\sqrt{12 - \sqrt{41 + 6\sqrt{32}}}$.
(10 points)

Question 9. To solve the radical equation $x^2 - 3x - \sqrt{x^2 - 3x + 2} = 0$. (10 points)

Question 10. To find the solution set of the exponential equation.

$$5^{4x} + 25 - 5^{3x+1} - 5^{x+1} = 0 \quad (10 \text{ points})$$

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Question 11. Given the exponential inequality $\left(\frac{1}{2}\right)^{x^2+2x+8} < \left(\frac{1}{4}\right)^{x+\frac{1}{2}}$. Find the solution set. (10 points)

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Question 12. Evaluate the value of logarithmic expression.

$$\log_{10} \sqrt{128} - 2 \log_{\frac{1}{10}} \sqrt{325} + \log_{\frac{1}{100}} 91 + 27^{(\log_3 2 + \log_3 81)} \quad (10 \text{ points})$$

Question 13. Given the logarithmic equation .

$$\log_{x^2} 3 + \log_{\sqrt{x}} 3 + \log_{\sqrt[3]{x}} 3 = 5\frac{1}{2}$$

Find the solution set. (10 points)

Question 14. Find all solutions of the logarithmic inequality.

$$\log_{\frac{1}{2}}(3x-4) > \log_{\frac{1}{2}}(x^2-2) \quad (10 \text{ points})$$

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Question 15. Find all possible solutions of a system of two exponential equations.

$$2^{x+y} = 6^y$$

$$3^{x-1} = 2^{y+1}$$

(10 points)

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